

Team ID: SWTID-2026-5919

Backend Fit Solution

Team ID: SWTID-2026-3342

Project Name: AI FAQ Assistant API

1. Problem

Many organizations receive the same frequently asked questions from customers and users. Managing these questions manually takes time and may result in delayed responses.

Common problems include:

- Users cannot get answers to common questions quickly.
- FAQ information is often scattered or difficult to organize.
- Administrators have to repeatedly answer the same questions.
- Manual customer support increases workload.
- Users may receive inconsistent answers.
- Searching through a large collection of FAQs can be time-consuming.

2. Target Customers / Users

The main users of the AI FAQ Assistant API are:

- Customers who need quick answers to common questions.
- Organizations that provide customer support.
- Educational institutions handling student queries.
- Small businesses and service providers.
- Administrators who manage FAQ information.

3. Existing Situation

Currently, users may depend on manually searching websites, reading FAQ pages, or contacting customer support to find answers.

This can lead to:

- Increased waiting time.
- Repeated questions to support staff.
- Difficulty in maintaining FAQ information.
- Higher workload for administrators.
- Poor user experience.

4. Proposed Solution

The AI FAQ Assistant API is a backend application that provides a centralized system for creating, managing, storing, and retrieving FAQ information.

The system is developed using **Node.js and Express.js** for REST API services and **MongoDB** for storing user accounts and FAQ records.

The API allows users to:

- Create FAQ records.
- Store and organize FAQ information.
- Retrieve frequently asked questions.
- Search for relevant FAQ content.
- Generate AI-based answers.
- Manage FAQ data through API endpoints.

5. Problem–Solution Fit

Customer Problem	Proposed Solution
Users wait for answers to common questions	AI-generated answers provide faster responses
FAQs are difficult to organize	Centralized FAQ management system
Support staff repeatedly answer similar questions	Automated FAQ assistance reduces repetitive work
Information is scattered	MongoDB stores FAQ information in one system
Searching FAQs takes time	API-based retrieval helps access relevant information
Users may receive inconsistent responses	Stored FAQ information provides standardized content

6. Customer Benefits

The AI FAQ Assistant API helps customers and organizations by:

- Reducing the time required to find answers.
- Providing quick responses to common questions.
- Reducing repetitive customer-support work.
- Organizing FAQ information efficiently.
- Improving accessibility of FAQ information.
- Supporting scalable backend services.
- Providing a foundation for AI-powered customer support.

7. Value Proposition

“An AI-powered FAQ Assistant API that helps organizations manage FAQ information and provide quick, consistent, and intelligent answers to frequently asked questions.”

8. How the Solution Fits the Customer

The solution addresses frequent customer-support problems by combining **FAQ management, database storage, REST APIs, and AI-generated responses** in one backend system.

Instead of depending completely on manual support, users can access relevant information through the API and receive AI-assisted answers.

This improves the existing FAQ process and creates a more convenient interaction between users and the organization.

9. Expected Outcome

The expected outcome of the project is to create an efficient and user-friendly backend system that can:

- Manage FAQ information.
 - Store user and FAQ data securely.
 - Retrieve relevant FAQ information.
 - Generate AI-assisted answers.
 - Reduce repetitive support activities.
 - Improve response time and user experience.
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10. Conclusion

The AI FAQ Assistant API provides a practical solution to the problem of managing and responding to frequently asked questions. By combining Node.js, Express.js, MongoDB, REST APIs, and AI-based response generation, the system provides a centralized approach for efficient FAQ management and automated assistance.